A Sparse Reconstruction Framework for Multi-Transmitter Radio Map Estimation

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***Abstract*—Radio map estimation has drawn increasing attention because of its broad utility in wireless communications and military operations. We focus on the *sampling-based* setting, where the target radio maps are estimated from sparse observations without access to the underlying transmitter realizations. We formulate this problem as a sparse reconstruction task for radio maps drawn from an underlying distribution. Under regularity assumptions, we show a decomposition of the expected population risk into an approximation term that scales as $\mathcal{O}(s^{-2/d})$ with the number of observed samples and an estimation term that depends on the number of training examples. We then empirically study how practical error rates compare with the scaling behavior predicted by the theory. Finally, we introduce RadioET, the first energy-based reconstruction method for radio map estimation. Experiments on a multi-transmitter radio map benchmark demonstrate the effectiveness of RadioET.**

***Index Terms*—Radio map estimation, sparse reconstruction, energy-based models.**

I. Introduction

Radio map estimation (RME) is the process of reconstructing quantities such as path loss or received signal strength across a geographic area from measurements at known locations [1]. Radio maps support network planning, resource allocation, and situational awareness [2]. Such applications are essential in emerging 6G and defense settings. Thus, it is critical to develop robust RME methods that generalize across diverse propagation environments.

We consider two RME scenarios depending on whether the transmitter (TX) realization is known [2]. In the *TX-conditioned* setting, the transmitter realization (the number of transmitters, their locations, and transmit powers) is modeled as part of the input. Consequently, the learning task reduces to approximating a deterministic function of the environment and transmitter parameters. In this paper, we focus on the *sampling-based* setting [3], where the transmitter realization is unknown. Unlike prior work that treats antenna locations as design variables to optimize system performance [4], [5], the transmitter geometry must be inferred indirectly from sparse measurements. In this case, the input consists of environment information and a sparse set of measurements.

In this work, we view this sparse reconstruction task through the lens of standard approximation and estimation error rates. Our contributions are summarized as follows:

- We formulate sampling-based *multi-transmitter* radio map estimation, where the transmitter count, locations, and powers are unknown and must be inferred indirectly from sparse measurements. We also construct a multi-transmitter benchmark from single-transmitter RadioMapSeer maps by superposing maps in power space.
- We develop an approximation–estimation analysis for sparse RME and derive an asymptotic reconstruction-risk bound with an $O(s^{-2/d})$ approximation term in the number of observed samples s over a d-dimensional domain, and a finite-sample estimation term that decays as $O((\frac{\log n}{n})^{\frac{1}{2}})$ with the number of training maps n. This provides a theoretical sparsity-scaling reference beyond the primarily numerical focus of most learning-based RME studies.
- We introduce RadioET, to our knowledge the first energy-based architecture for RME, and show that it consistently outperforms existing models on the proposed multi-transmitter benchmark, with empirical sparsity scaling consistent with the $O(1/s)$ reference rate in two dimensions.

The rest of this paper is organized as follows. Section II reviews related work on RME approaches. Sections III and IV formalize the problem and present the theoretical analysis. Section V describes the architecture of RadioET. Section VI presents experimental results on the multi-transmitter RME benchmark. Finally, Section VII concludes the paper with a discussion of limitations and future directions.

II. Related Work

Early data-driven RME approaches often used spatial interpolation and geostatistical methods, such as Kriging [1] and inverse distance weighting (IDW) [6]. These methods reconstruct radio maps from sparse observations by assuming spatial correlation or smooth distance-based variation. However, they typically do not explicitly model environmental structure or propagation physics, which can limit accuracy in obstructed and complex settings.

Deep learning has recently become a common approach to RME. In the TX-conditioned scenario, representative architectures include convolutional encoder-decoder models and generative methods such as RadioUNet [7], RadioMamba [8], RadioDiff [9], and RME-GAN [10]; and physics-based approaches such as PhyRMDM [11] and RadioDUN [12]. Joint reconstruction has also been explored, where UAV measurements are used to estimate both the radio and environment maps [13]. Related work has investigated the sampling-based RME problem under extreme levels of sparsity. For example, DAT-Unet [3] uses deformable attention with neighborhood attention to efficiently process sparse measurements.

Beyond RME, sparse reconstruction has been studied in other domains. Recent work includes REPA [14] for reconstructing sparsely observed vector fields using self-supervised learning, and the Energy Transformer [15], [16] for reconstructing flow fields through energy minimization. While these methods demonstrate the effectiveness of learned priors for sparse reconstruction, energy-based approaches have not yet been explored for radio map estimation.

III. Problem Formulation

A. Preliminaries

Let $D = [0,1]^d$ denote the spatial domain, taken to be the unit cube so that its volume is $|D| = 1$. For the discrete analysis below, partition D into $N = L^d$ identical cubic cells with side length $1/L$ and volume $1/N$. Let $\Lambda = \{c_1, \dots, c_N\}$ denote their centers. Let $\mathcal{B}$ denote the environmental information available to the estimator (e.g. buildings, cars). A transmitter configuration is denoted by $C = \big(m, \{(t_j, p_j)\}_{j=1}^m\big)$, where m is the number of active transmitters, $t_j \in \Lambda$ is the location of transmitter j, and $p_j > 0$ is its linear transmit-power scale.

Let g be a deterministic radio wave propagation operator from transmitter configurations and environment maps to radio maps on this grid. In the TX-conditioned setting, C and $\mathcal{B}$ are both part of the input, and the conditional law is the Dirac mass $\mathcal{P}_{F|C,\mathcal{B}}(\cdot \mid c, b) = \delta_{g(c,b)}$, where F denotes a random map. In other words, conditioned on the full input, there is no non-degenerate distribution over target maps and the learning problem reduces to deterministic operator regression.

In the sampling-based setting, the transmitter configuration is unobserved. The estimator is provided with the environment $\mathcal{B}$, but not the configuration C. A random configuration C and environment $\mathcal{B}$ are drawn and passed through g to generate a radio map $F = g(C, \mathcal{B})$. We denote by $F \sim \mathcal{P}$ the resulting distribution over radio maps induced by the randomness in C and $\mathcal{B}$. Since the transmitter configuration is not observed by the estimator, the target map remains stochastic even after conditioning on the available inputs.

B. Teacher–student formalization

The *teacher* randomly samples maps from the distribution $\mathcal{P}$. A random map is a function $F : \Lambda \to \mathbb{R}^k$, where $k = 1$ for scalar radio maps. For a fixed sparsity level s, observation locations $X_1, \dots, X_s$ are drawn independently and uniformly on Λ ($s \ll N$), and the sparse observation tuple is

$$O_s = \big((X_1, Y_1), \dots, (X_s, Y_s)\big),$$

where $Y_j = F(X_j)$ is the observed value at location X_j.

A *student* reconstructor with parameters $\theta \in \Theta$ is a network

$$\widehat{F}_\theta(c; \mathcal{B}, O_s) : \Lambda \times \mathcal{B} \times (\Lambda \times \mathbb{R}^k)^s \to \mathbb{R}^k,$$

giving its prediction at a query cell c. Evaluating $\widehat{F}_\theta(\mathcal{B}, O_s)$ at all cells yields the reconstructed map $\widehat{F}_\theta(\mathcal{B}, O_s)$. For two maps $f, \widetilde{f} : \Lambda \to \mathbb{R}^k$, the reconstruction loss is the mean squared error

$$\ell(f, \widetilde{f}) = \frac{1}{N} \sum_{c \in \Lambda} \|f(c) - \widetilde{f}(c)\|_2^2.$$

Accordingly, the population risk of a reconstructor $\widehat{F}_\theta$ is

$$\mathcal{L}_{\text{gen}}^{(s)}(\widehat{F}_\theta) = \mathbb{E}\Big[\ell\big(F, \widehat{F}_\theta(\mathcal{B}, O_s)\big)\Big],$$

and the best-in-class risk is

$$\delta(s) = \inf_{\theta \in \Theta} \mathcal{L}_{\text{gen}}^{(s)}(\widehat{F}_\theta).$$

Given n training examples $T_s^{(i)} = (\mathcal{B}^{(i)}, O_s^{(i)}, F^{(i)})$, the empirical risk is

$$\mathcal{L}_n^{(s)}(\widehat{F}_\theta) = \frac{1}{n} \sum_{i=1}^n \ell\big(F^{(i)}, \widehat{F}_\theta(\mathcal{B}^{(i)}, O_s^{(i)})\big),$$

and an empirical risk minimizer is $\widehat{F}_{\theta^*_{n,s}}$, where

$$\theta^*_{n,s} \in \arg\min_{\theta \in \Theta} \mathcal{L}_n^{(s)}(\widehat{F}_\theta).$$

The expected population risk of the ERM is

$$\delta(n, s) = \mathbb{E}\big[\mathcal{L}_{\text{gen}}^{(s)}(\widehat{F}_{\theta^*_{n,s}})\big],$$

where the outer expectation is over the random training set.

IV. Theoretical Analysis

In this section, we give the complete statements and proofs of the sample-complexity results for the sparse-reconstruction formalization on the discrete L^d grid. These results specialize classical bounds from nonparametric regression [17] and statistical learning theory [18] to our discretized setting. To keep the notation light, we will drop $\mathcal{B}$, and accordingly we write $\widehat{F}_\theta(O_s)$ for $\widehat{F}_\theta(\mathcal{B}, O_s)$, and a training example becomes $T_s = (O_s, F)$. As $k = 1$ for scalar radio maps, fields are real-valued throughout.

We take the student class $\Theta = \Theta_{s,d,L}$ to be the *bounded-output ReLU query class*: each $\widehat{F}_\theta \colon \Lambda \times (\Lambda \times \mathbb{R})^s \to \mathbb{R}$ is a feedforward ReLU network whose final two units apply the clamp

$$\text{clip}_B(z) = \sigma(z+B) - \sigma(z-B) - B, \qquad \sigma(z) = \max\{z, 0\},$$

which maps $\mathbb{R}$ onto $[-B, B]$.

Introducing a query point $X \sim \text{Unif}(\Lambda)$ independent of everything else, the cell average in the loss is an expectation, so the population risk admits the equivalent form

$$\mathcal{L}_{\text{gen}}^{(s)}(\widehat{F}_\theta) = \mathbb{E}\big[(F(X) - \widehat{F}_\theta(X; O_s))^2\big],$$

where the expectation is over F, O_s, and X. The induced loss class is $\mathcal{G}_s = \{T_s \mapsto g_\theta(T_s) : \theta \in \Theta\}$, where $g_\theta(T_s) = \ell\big(F, \widehat{F}_\theta(O_s)\big)$.

The results of this section rely on the following assumptions:

1) For each fixed s, the sample locations $X_1, \dots, X_s$ are i.i.d. uniform on Λ.

2) Every map drawn by the teacher is L_0-Lipschitz on the grid, i.e. $|F(c) - F(c')| \le L_0\|c - c'\|_2$ for all $c, c' \in \Lambda$, with L_0 independent of L. Equivalently, F is the restriction to Λ of an L_0-Lipschitz field on D.
3) Teacher and student maps are uniformly bounded, i.e. $|F(c)| \le B$ and $|\widehat{F}_\theta(c; O_s)| \le B$ for all $c \in \Lambda$, all $\theta \in \Theta$, and all observation tuples.
4) For fixed s, the induced loss class $\mathcal{G}_s$ has finite pseudo-dimension, $\mathrm{Pdim}(\mathcal{G}_s) = d_s < \infty$.

Definition 1 (Nearest-neighbor reconstructor) *For a cell $c \in \Lambda$ and observations O_s, write $R_c = \min_{1\le j\le s}\|c - X_j\|_2$ for the distance to the nearest sample and $\mathcal{N}(c) = \{\, j : \|c - X_j\|_2 = R_c \,\}$ for the set of nearest samples. The nearest-neighbor reconstructor predicts the average of the observed values there,*

$$\widehat{F}_{\mathrm{NN}}(c; O_s) \;=\; \frac{1}{|\mathcal{N}(c)|}\sum_{j\in\mathcal{N}(c)} Y_j .$$

Lemma 1 (Rate for the NN reconstructor) *For $d \ge 2$,*

$$\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\mathrm{NN}}) \;\le\; C_1 L_0^2\, s^{-2/d},$$

where C_1 is the constant in the nearest-neighbor distance bound of [17, Theorem 2.1].

Proof: Fix c and a realization of O_s. Every $j \in \mathcal{N}(c)$ has $\|c - X_j\|_2 = R_c$, so

$$\begin{aligned}\big|F(c) - \widehat{F}_{\mathrm{NN}}(c; O_s)\big| &= \Big|\tfrac{1}{|\mathcal{N}(c)|}\textstyle\sum_{j\in\mathcal{N}(c)}\big(F(c) - Y_j\big)\Big| \\ &\le \tfrac{1}{|\mathcal{N}(c)|}\textstyle\sum_{j\in\mathcal{N}(c)}\big|F(c) - F(X_j)\big| \\ &\le L_0 R_c,\end{aligned}$$

using $Y_j = F(X_j)$ and Assumption 2. Both sides are nonnegative, so squaring gives $\big(F(c) - \widehat{F}_{\mathrm{NN}}(c; O_s)\big)^2 \le L_0^2 R_c^2$ for every $c \in \Lambda$ and every realization of O_s. By the definition of $\mathcal{L}^{(s)}_{\mathrm{gen}}$ and linearity of expectation over the finite sum,

$$\begin{aligned}\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\mathrm{NN}}) &= \mathbb{E}\Big[\tfrac{1}{N}\textstyle\sum_{c\in\Lambda}\big(F(c) - \widehat{F}_{\mathrm{NN}}(c; O_s)\big)^2\Big] \\ &\le \frac{L_0^2}{N}\sum_{c\in\Lambda}\mathbb{E}\big[R_c^2\big].\end{aligned}$$

But

$$\frac{1}{N}\sum_{c\in\Lambda}\mathbb{E}\big[R_c^2\big] = \mathbb{E}\Big[\min_{1\le j\le s}\|c - X_j\|_2^2\Big]$$

is the expected squared nearest-neighbor distance under uniform sampling, which [17, Theorem 2.1] bounds by $C_1 s^{-2/d}$ for $d \ge 2$. Substituting gives $\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\mathrm{NN}}) \le C_1 L_0^2 s^{-2/d}$. ■

Exact computation of the NN map with ReLUs. By Lemma 1 it suffices that $\widehat{F}_{\mathrm{NN}}$ lie in the student class, and on the finite grid it does so exactly. A ReLU network only forms weighted sums and applies $\sigma(z) = \max\{z, 0\}$, so it cannot directly multiply two inputs, square, take a minimum, or divide. Yet the nearest-neighbor map needs all of these. Each is still rebuilt from σ and additions, and the rebuilt versions are exact because the inputs are grid points. Every coordinate gap is a multiple of $1/L$ and every squared distance a multiple of $1/L^2$, so each operation need only be correct at these finitely many values.

Proposition 1 (Exact ReLU realization) *Write $\sigma(z) = \max\{z, 0\}$. There is a feedforward ReLU network $\widehat{F}_\star$, using $O(sdL)$ units such that for every query $x \in \Lambda$ and every observation tuple O_s with $X_j \in \Lambda$ and $Y_j \in [-B, B]$,*

$$\widehat{F}_\star(x; O_s) \;=\; \widehat{F}_{\mathrm{NN}}(x; O_s).$$

Proof: We construct $\widehat{F}_\star$ explicitly. Each coordinate of a grid point lies in $\{(a + \frac{1}{2})/L : 0 \le a \le L - 1\}$, so every coordinate difference is an integer multiple of $1/L$ and its absolute value lies in $\{0, \frac{1}{L}, \dots, \frac{L-1}{L}\}$. The six primitives below each rebuild one operation the map needs out of σ and weighted sums:

$$\begin{aligned}\mathrm{abs}(t) &= \sigma(t) + \sigma(-t),\\ \mathrm{sq}(u) &= \frac{u}{L} + \frac{2}{L}\sum_{m=1}^{L-1}\sigma\Big(u - \tfrac{m}{L}\Big),\\ \min(a, b) &= a - \sigma(a - b),\\ \mathrm{flag}(g) &= \sigma\big(1 - L^2 g\big),\\ \mathrm{gate}_M(f, v) &= \sigma\big(v + 2M(f - 1)\big)\\ &\quad - \sigma\big(-v + 2M(f - 1)\big),\\ \mathrm{eq}_k(K) &= \sigma\big(1 - \mathrm{abs}(K - k)\big).\end{aligned}$$

Here abs returns the size of a coordinate gap, and sq gives the exact square of its input at every possible value $u \in \{0, \frac{1}{L}, \dots, \frac{L-1}{L}\}$, so $\mathrm{sq}(u) = u^2$ there. The map min returns the smaller of two values, $\min(a, b) = \min\{a, b\}$. The flag $\mathrm{flag}(g)$ equals 1 when $g = 0$ and 0 once $g \ge 1/L^2$. The switch gate_M, valid for $f \in \{0, 1\}$ and $|v| \le M$, returns v if $f = 1$ and 0 if $f = 0$. The test eq_k returns 1 if the integer K equals k and 0 otherwise. Two of these are particularly needed for operations σ cannot perform directly. A ReLU network cannot multiply two of its own inputs, so it cannot form the product $f_j Y_j$ that keeps the winner's value. Once f_j is known to be 0 or 1, however, gate reproduces $f_j Y_j$ exactly. A ReLU network also cannot divide by a quantity it has computed, which is what the final average S/K needs. Since the count K is one of the integers $1, \dots, s$, eq tests K against each candidate k and the network reads off S/k at the unique match, turning division by an unknown integer into a finite selection. The primitives chain into the natural pipeline, computed in sequence:

(A) squared distances $d_j = \sum_{i=1}^{d}\mathrm{sq}\big(\mathrm{abs}(x^{(i)} - X_j^{(i)})\big)$, $1 \le j \le s$;
(B) the nearest squared distance $R = \min(d_1, \dots, d_s)$,
(C) flags $f_j = \mathrm{flag}(d_j - R)$, $1 \le j \le s$;
(D) the gated sum $S = \sum_{j=1}^{s}\mathrm{gate}_B(f_j, Y_j)$ and the count $K = \sum_{j=1}^{s} f_j$;
(E) the output $\widehat{F}_\star(x; O_s) = \sum_{k=1}^{s}\mathrm{gate}_{sB}\big(\mathrm{eq}_k(K),\, S/k\big)$.

Exactness on Λ follows in three steps. (i) Each coordinate gap $x^{(i)} - X_j^{(i)}$ is a multiple of $1/L$, so abs returns the gap's size exactly and sq its exact square. Summing over the coordinates, step (A) returns the true squared distance

$d_j = \|x - X_j\|_2^2$, an integer multiple of $1/L^2$, and step (B) returns the minimum $R = \min_j d_j$. (ii) Because every d_j and R is an integer multiple of $1/L^2$, so is $d_j - R \geq 0$, which is therefore either 0 (exactly when j is a nearest sample) or at least $1/L^2$. The flag in step (C) sends these two cases to 1 and 0, so $f_j = \mathbf{1}\{j \in \mathcal{N}(x)\}$ marks the nearest samples and $K = \sum_j f_j = |\mathcal{N}(x)|$ counts them, an integer in $\{1, \ldots, s\}$. (iii) Each f_j is 0 or 1 and $|Y_j| \leq B$, so the gates in step (D) are exact and keep only the nearest samples, giving the sum $S = \sum_{j \in \mathcal{N}(x)} Y_j$ and the count K, with $|S| \leq sB$. Because K is an integer in $\{1, \ldots, s\}$, exactly one test $\mathrm{eq}_k(K)$ equals 1, the one with $k = K$, and $|S/k| \leq sB$, so the step (E) gates select S/k at $k = K$ and return S/K. This is the average of the nearest samples' values, so $\widehat{F}_\star(x; O_s) = S/K = \widehat{F}_{\mathrm{NN}}(x; O_s)$. ■

Remark 1 (Size of the network) *Counting computing-units one by one (each ReLU operation adds one):* abs *uses* 2*,* sq *uses* $L - 1$*,* min *and* flag *use* 1 *each,* gate *uses* 2*, and* eq *uses* 3 *(an* abs *followed by one more* σ*). The five blocks then contribute*

$$\begin{aligned}&\underbrace{sd\big(2 + (L-1)\big)}_{(A)} + \underbrace{(s-1)}_{(B)} + \underbrace{s}_{(C)} + \underbrace{2s}_{(D)} + \underbrace{5s}_{(E)} \\ &\quad = sd\,(L+1) + 9s - 1 \\ &\quad = \mathcal{O}(sdL).\end{aligned}$$

Theorem 1 (Approximation error) *If the student class* $\Theta_{s,d,L}$ *contains the network* $\widehat{F}_\star$ *of Proposition 1, then, writing* $C_{\mathrm{NN}} := C_1 L_0^2$ *(independent of* s*),*

$$\delta(s) \;\leq\; C_{\mathrm{NN}}\, s^{-2/d}. \tag{1}$$

Proof: By Proposition 1, $\widehat{F}_\star \in \Theta_{s,d,L}$ satisfies $\widehat{F}_\star(x; O_s) = \widehat{F}_{\mathrm{NN}}(x; O_s)$ for every $x \in \Lambda$ and every O_s. Thus, the two reconstructors therefore incur the same loss on every realization of (F, O_s), so $\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_\star) = \mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\mathrm{NN}})$. Since $\delta(s) = \inf_{\theta \in \Theta_{s,d,L}} \mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_\theta) \leq \mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_\star)$, Lemma 1 bounds the right-hand side by $C_1 L_0^2 s^{-2/d} = C_{\mathrm{NN}} s^{-2/d}$. ■

Throughout, $\mathrm{Pdim}(\mathcal{G}_s) = d_s < \infty$ by Assumption 4, and Assumption 3 gives the per-example bound

$$\begin{aligned}0 \leq g_\theta(T_s) &= \frac{1}{N}\sum_{c \in \Lambda}\big(F(c) - \widehat{F}_\theta(c; O_s)\big)^2 \\ &\leq \frac{1}{N}\sum_{c \in \Lambda}(2B)^2 = 4B^2 \;=:\; M.\end{aligned}$$

Lemma 2 (Two-sided uniform convergence) *With* $0 \leq g_\theta \leq M$ *and* $\mathrm{Pdim}(\mathcal{G}_s) = d_s < \infty$*, for every* $\eta \in (0,1)$*, with probability at least* $1 - \eta$*, simultaneously for all* $\theta \in \Theta$*,*

$$\begin{aligned}\big|\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_\theta) - \mathcal{L}^{(s)}_n(\widehat{F}_\theta)\big| &\leq M\sqrt{\frac{2d_s \log(en/d_s)}{n}} \\ &\quad + M\sqrt{\frac{\log(1/\eta)}{2n}}.\end{aligned}$$

Proof: This is the pseudo-dimension uniform-convergence bound for a class of $[0, M]$-valued functions [18, Theorem 11.8], applied to $\mathcal{G}_s$, which has range $[0, M]$ and pseudo-dimension d_s. ■

Write $\varepsilon_n(\eta) := M\sqrt{\dfrac{2d_s \log(en/d_s)}{n}} + M\sqrt{\dfrac{\log(1/\eta)}{2n}}$.

Proposition 2 (High-probability ERM excess risk) *For every* $\eta \in (0,1)$*, with probability at least* $1 - \eta$*,*

$$\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\theta^*_{n,s}}) - \delta(s) \;\leq\; 2\varepsilon_n(\eta).$$

Proof: By Lemma 2, with probability at least $1 - \eta$ we have $\big|\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_\theta) - \mathcal{L}^{(s)}_n(\widehat{F}_\theta)\big| \leq \varepsilon_n(\eta)$ simultaneously for all $\theta \in \Theta_{s,d,L}$. On that event, for every θ,

$$\begin{aligned}\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\theta^*_{n,s}}) &\leq \mathcal{L}^{(s)}_n(\widehat{F}_{\theta^*_{n,s}}) + \varepsilon_n(\eta) \\ &\leq \mathcal{L}^{(s)}_n(\widehat{F}_\theta) + \varepsilon_n(\eta) \\ &\leq \mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_\theta) + 2\varepsilon_n(\eta),\end{aligned}$$

where the first and third inequalities apply the two-sided bound to $\theta^*_{n,s}$ and to θ respectively, and the middle inequality is the empirical optimality of $\theta^*_{n,s}$. This holds for every $\theta \in \Theta_{s,d,L}$, so taking the infimum over θ and using $\delta(s) = \inf_\theta \mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_\theta)$ gives the stated bound. ■

Theorem 2 (Estimation error) *For every* $\eta \in (0,1)$*,*

$$\delta(n,s) - \delta(s) \;\leq\; 2\varepsilon_n(\eta) + \eta M.$$

In particular, choosing $\eta = 1/n$*,*

$$\begin{aligned}\delta(n,s) - \delta(s) &\leq 2M\sqrt{\frac{2d_s \log(en/d_s)}{n}} \\ &\quad + 2M\sqrt{\frac{\log n}{2n}} + \frac{M}{n}.\end{aligned}$$

With $M = 4B^2$ *and* $d_s < \infty$ *(Assumption 4), for fixed* s*,*

$$\delta(n,s) - \delta(s) \;=\; O\left(B^2\sqrt{\frac{d_s \log n}{n}}\right). \tag{2}$$

Proof: Let $\Delta_n := \mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\theta^*_{n,s}}) - \delta(s)$. By Proposition 2, $\mathbb{P}\big(\Delta_n \leq 2\varepsilon_n(\eta)\big) \geq 1 - \eta$. Since $0 \leq g_\theta \leq M$ we have $0 \leq \delta(s) \leq M$ and $0 \leq \mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\theta^*_{n,s}}) \leq M$, and $\mathcal{L}^{(s)}_{\mathrm{gen}}(\widehat{F}_{\theta^*_{n,s}}) \geq \delta(s)$, so $0 \leq \Delta_n \leq M$. Hence

$$\begin{aligned}\mathbb{E}[\Delta_n] &\leq 2\varepsilon_n(\eta)\,\mathbb{P}\big(\Delta_n \leq 2\varepsilon_n(\eta)\big) \\ &\quad + M\,\mathbb{P}\big(\Delta_n > 2\varepsilon_n(\eta)\big) \leq 2\varepsilon_n(\eta) + \eta M.\end{aligned}$$

Since $\mathbb{E}[\Delta_n] = \delta(n,s) - \delta(s)$, this is the first bound. For the second, set $\eta = 1/n$. Then $\log(1/\eta) = \log n$, so $\varepsilon_n(\eta) = M\sqrt{2d_s \log(en/d_s)/n} + M\sqrt{\log n/(2n)}$ and $\eta M = M/n$, and substituting into $\delta(n,s) - \delta(s) \leq 2\varepsilon_n(\eta) + \eta M$ gives the second bound. Finally $d_s \geq 1$ gives $\log(en/d_s) \leq \log(en) = 1 + \log n = O(\log n)$, and the $\sqrt{\log n/(2n)}$ and $1/n$ terms are of lower order than $\sqrt{d_s \log(en/d_s)/n}$. With $M = 4B^2$, $\delta(n,s) - \delta(s) = O\big(B^2\sqrt{d_s \log n/n}\big)$. ■

Corollary 1 (Total reconstruction error) *There exist constants* $C_{\mathrm{NN}} > 0$ *and* $C_{\mathrm{E}} > 0$*, such that for fixed* s*,*

$$\delta(n,s) \;\leq\; \underbrace{C_{\mathrm{NN}}\, s^{-2/d}}_{\textit{approximation}} + \underbrace{C_{\mathrm{E}}\sqrt{\tfrac{d_s \log n}{n}}}_{\textit{estimation}},$$

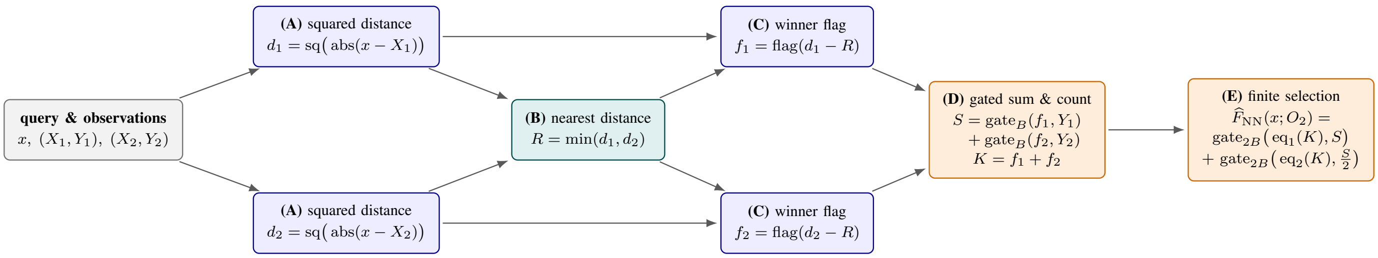

Fig. 1. Toy ReLU network ($d = 1$, $s = 2$) that realizes the nearest-neighbor reconstructor *exactly*. Data flow left to right through the pipeline of Proposition 1. Per-sample stages (blue) compute a squared distance (A) and a winner flag (C), the reduce stage (teal) takes the nearest distance (B), and the combine stages (orange) form the gated sum and count (D) and read off the average by finite selection (E).

up to lower-order terms in n. For the $L \times L$ grid ($d = 2$),

$$\delta(n, s) \ \le\ \frac{C_{\mathrm{NN}}}{s} + C_{\mathrm{E}}\sqrt{\frac{d_s \log n}{n}}.$$

Proof: Decompose $\delta(n, s) = \delta(s) + \big(\delta(n, s) - \delta(s)\big)$. Theorem 1 bounds the first term by $C_{\mathrm{NN}}\, s^{-2/d}$ with $C_{\mathrm{NN}} = C_1 L_0^2$; and Theorem 2 bounds the second, $\delta(n, s) - \delta(s) = O\big(B^2 \sqrt{d_s \log n / n}\big)$, which we write as $C_{\mathrm{E}}\sqrt{d_s \log n / n}$ up to lower-order terms in n, with C_{E} absorbing B^2 and the remaining constants. Adding the two bounds and substituting $d = 2$ gives the bound. ∎

The first term is the approximation error, bounded in (1), caused by observing only s cells. The second term is the estimation error, bounded in (2), due to learning the reconstructor using a finite dataset of n examples. For a two-dimensional domain $d = 2$, our bound predicts that the approximation error decays at least inversely with the number of observed samples.

In practice, the assumptions underlying this rate may only be approximately satisfied in RME settings. For example, sampling may be non-uniform since measurements are often unavailable inside buildings, and radio maps may violate smoothness assumptions due to sharp transitions at building boundaries and shadowing effects. Whether this rate describes how practical reconstruction error decays when these assumptions are relaxed is an empirical question, which we study next.

V. Model Architecture

In this section, we propose RadioET, an energy-based reconstruction model based on ConvNeXt layers [19] and the Energy Transformer [15]. Both the encoder and decoder use hierarchical convolutional layers. The encoder extracts local features and reduces the spatial resolution of the input at each stage. At the bottleneck, Energy Transformer blocks model global spatial relationships and estimate missing regions from sparse observations. The decoder reconstructs the output by combining encoder features through skip connections. Fig. 2 shows the overall architecture.

The Energy Transformer [15] uses a single recurrent block that iteratively updates its tokens to minimize a global energy term. Let $x_C \in \mathbb{R}^D$ be the token at spatial location C and g_C its layer-normalized version. Throughout, B, C index the N tokens and $j = 1, \ldots, D$ runs over feature coordinates. The block defines two separate energy terms. The first is the attention energy,

$$E^{\mathrm{ATT}} = -\frac{1}{\beta}\sum_{h=1}^{H}\sum_{C}\log\Bigg(\sum_{B \neq C}\exp\big(\beta\, A_{hBC}\big)\Bigg), \tag{3}$$

where the score $A_{hBC} = \sum_a K_{ahB}\, Q_{ahC}$ couples tokens B and C within attention head h, built from keys $K_{ahB} = \sum_j W^K_{ahj} g_{jB}$ and queries $Q_{ahC} = \sum_j W^Q_{ahj} g_{jC}$. Here W^K, W^Q are learnable projections, a indexes the per-head attention channels, and β is an inverse temperature [15].

The second is the Hopfield (associative-memory) energy that pulls each token toward plausible patterns,

$$E^{\mathrm{HN}} = -\sum_{B}\sum_{\mu=1}^{M} G\Big(\sum_j \xi_{\mu j}\, g_{jB}\Big), \tag{4}$$

where the M rows of ξ are learnable memory patterns and G is a smooth nonlinearity whose derivative $G' = r$ is the activation function. The total global energy at time step t, denoted E^t, is the sum of the terms given in (3) and (4). The tokens are evolved by gradient descent on this energy for a total of T steps with step size α [15]:

$$x^{t+1} = x^t - \alpha\, \nabla_g E^t, \qquad t = 1, \ldots, T.$$

Lastly, the choices of architectural parameters we employed for RadioET are given in Table I.

TABLE I
RadioET Architectural Parameters

Parameter	Value	Parameter	Value
Stage dims	48, 96, 192, 384	Number of heads	8
Encoder blocks	2, 2, 2, 4	Hopfield r	Softmax
Decoder blocks	2, 2, 2, 2	Attention β	$\frac{1}{\sqrt{64}}$
Bottleneck dim	512	Time steps T	12
QK dim	64	Step size α	0.1

VI. Experiments

We conduct two main experiments using the RadioMapSeer dataset [20]. To expand the sampling space and introduce greater stochastic variation, we construct a multi-transmitter RME benchmark.

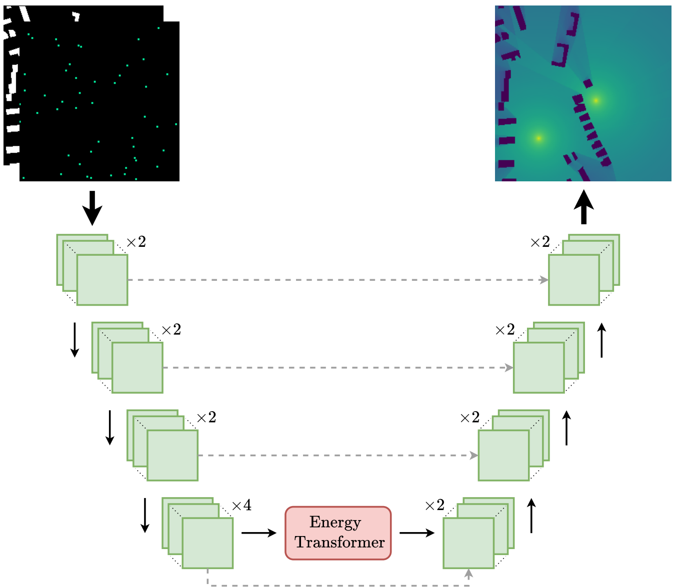

Fig. 2. Architecture of RadioET. Dashed lines indicate skip connections.

A. *Baseline Comparisons*

In order to create a realistic and challenging multi-transmitter RME benchmark, we superpose pathloss maps from the RadioMapSeer dataset [20] in linear power space. Specifically, we randomize the number, locations, and powers of the transmitters. Let $\mathcal{N}_{\text{tx}}$ denote the set of possible transmitter counts and let $\mathcal{M}_{\text{tx}}$ denote the transmit power-offset range in dB. Because weaker signals are truncated to -147 dB in the original dataset, sentinel pixels at this floor value contribute no linear power to the mixture. After synthesis, all outputs are placed on a common path loss scale $[0, 1]$, where the lower end corresponds to the RadioMapSeer floor of -147 dB and the upper end corresponds to the worst-case aggregate value of $-47 + \max \mathcal{M}_{\text{tx}} + 10 \log_{10}(\max \mathcal{N}_{\text{tx}})$. Using the procedure outlined in Algorithm 1, we generate random training examples on the fly during training.

We create dataset splits by randomly assigning 501, 100, and 100 city layouts to the training, validation, and test sets, respectively. We use the radio maps generated by DPM simulations from the RadioMapSeer dataset. In this experiment, we vary the number of observed samples (s) uniformly within $[10, 100]$ during training to evaluate the models' flexibility under varying degrees of sparsity. This corresponds to sampling between 0.015% and 0.15% of the spatial domain. We also include the car objects in the inputs to introduce additional obstructions.

We compare RadioET against DAT-Unet [3] and RadioUNet [7], which we reimplemented in PyTorch Lightning to ensure identical training pipelines. During validation and evaluation phases, we calculate RMSE and NMSE only on unobserved free-space pixels. For PSNR and SSIM, predictions are overwritten with ground truth values at observed pixels and zeros inside buildings. For evaluation metrics, we clamp

Algorithm 1 Multi-transmitter map synthesis

Input: Single-TX path loss maps $\{L_k\}_{k=1}^{80}$ in dB, count support $\mathcal{N}_{\text{tx}}$, power-offset range $\mathcal{M}_{\text{tx}}$

$n \sim \mathcal{U}(\mathcal{N}_{\text{tx}})$ {number of active transmitters}
sample distinct indices $\mathcal{I} = \{i_1, \ldots, i_n\}$ uniformly from $\{1, \ldots, 80\}$
$T(x) \leftarrow 0$ for all pixels x
for each $i \in \mathcal{I}$ **do**
  $m_i \sim \mathcal{U}(\mathcal{M}_{\text{tx}})$ {power offset in dB}
  $\alpha_i \leftarrow 10^{m_i/10}$ {linear power scale}
  $M_i(x) \leftarrow \mathbf{1}_{\{L_i(x) > -147\}}$
  $T(x) \leftarrow T(x) + \alpha_i \, M_i(x) \, 10^{L_i(x)/10}$
end for
$L(x) \leftarrow \begin{cases} \max\{10 \log_{10} T(x), -147\}, & T(x) > 0, \\ -147, & T(x) = 0 \end{cases}$
return L

predictions to $[0, 1]$.

All models use the same training setup given in Table III. Since RadioUNet consists of two training stages, we train both stages for 150 epochs. Additionally, we perform a lightweight hyperparameter search on the validation set. We perform a grid search over learning rates. For DAT-Unet, we additionally tune the drop path rate. Specifically, DAT-Unet is evaluated on the learning rates $\{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$ and drop path rates $\{0.0, 0.1, 0.2\}$. RadioET is evaluated using learning rates $\{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$ and RadioUNet using learning rates $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$. All models are trained with a masked MSE loss computed only on unobserved and free-space pixels. The inputs to the models consist of the observed path loss values and the binary observation, building, and car masks.

We train each model using 3 random seeds and report the mean and standard deviation of the evaluation metrics averaged on the test set. The results under varying sparsity regimes are summarized in Table II. In all metrics and sparsity regimes, the performance gains of RadioET are consistent. We also provide qualitative comparisons in Fig. 3.

B. *Empirical sparsity scaling*

We study how reconstruction error scales with the number of observed samples and compare the empirical trend to the upper bound on $\delta(s) = \mathcal{O}(1/s)$ in Section IV. Separate RadioET models are trained at sparsities $s \in \{10, 20, 40, 80, 160\}$ with a single seed since variance is small (Table II). To better align with the reconstruction loss used in our analysis, MSE is computed over all pixels. The number of training examples n is kept fixed through the number of city layouts in the training split, while Algorithm 1 generates different multi-transmitter maps. This isolates the effect of the number of observed samples per map. Furthermore, the grid dimension is fixed at 256×256 ($L = 256$, $d = 2$).

Since $\delta(s)$ is a population best-in-class risk, it cannot be measured directly from a finite dataset, so we instead bracket it. Let θ^* attain the infimum in $\delta(s)$. Because the ERM

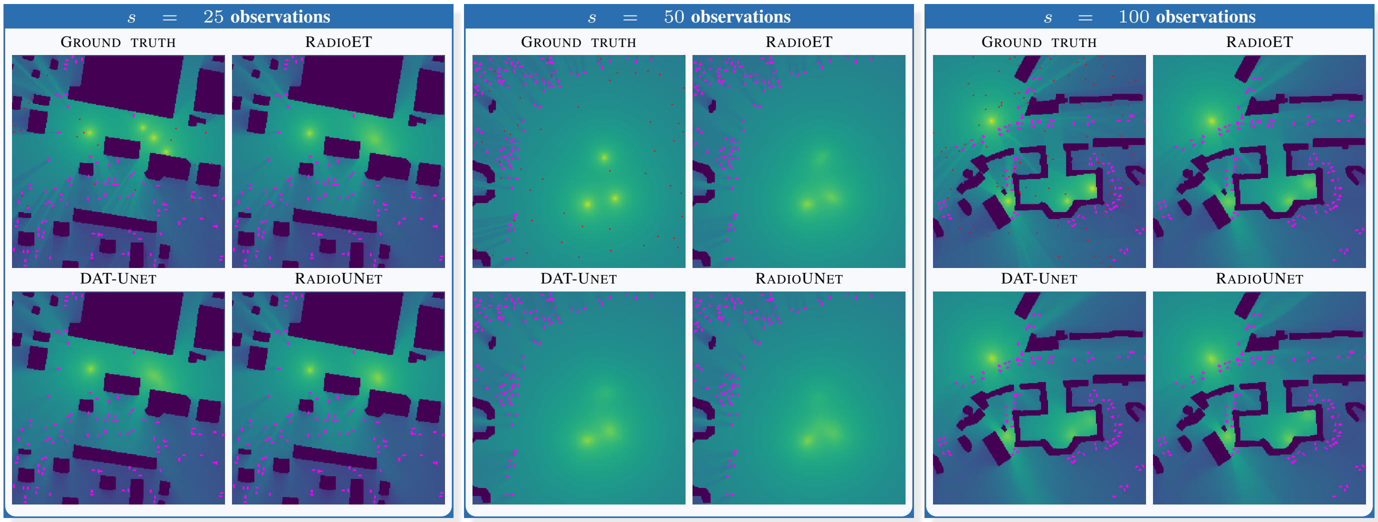

Fig. 3. Qualitative comparisons for multi-transmitter RME at $s = 25$, 50, and 100 observed samples (red points). Purple and pink blocks denote buildings and cars, respectively.

TABLE II
RESULTS ON THE TEST-SET. **BOLD** MARKS THE BEST METHOD.

$s \in$	Model	RMSE ↓	NMSE ↓	SSIM ↑	PSNR ↑
[10, 25]	DAT-Unet	0.03814 ± 0.00013	0.00814 ± 0.00006	0.9553 ± 0.0001	30.55 ± 0.03
	RadioUNet	0.03862 ± 0.00016	0.00834 ± 0.00007	0.9588 ± 0.0002	30.59 ± 0.04
	RadioET	0.03652 ± 0.00046	0.00746 ± 0.00019	0.9599 ± 0.0009	31.06 ± 0.14
[25, 50]	DAT-Unet	0.02700 ± 0.00006	0.00408 ± 0.00002	0.9623 ± 0.0001	33.17 ± 0.02
	RadioUNet	0.02635 ± 0.00017	0.00388 ± 0.00005	0.9666 ± 0.0003	33.54 ± 0.07
	RadioET	0.02438 ± 0.00052	0.00333 ± 0.00014	0.9684 ± 0.0011	34.16 ± 0.21
[50, 100]	DAT-Unet	0.02223 ± 0.00006	0.00277 ± 0.00001	0.9657 ± 0.0000	34.62 ± 0.02
	RadioUNet	0.02060 ± 0.0002	0.00238 ± 0.00005	0.9707 ± 0.0003	35.36 ± 0.09
	RadioET	0.01923 ± 0.0005	0.00207 ± 0.00011	0.9726 ± 0.0011	35.93 ± 0.25
[10, 100]	DAT-Unet	0.02695 ± 0.00005	0.00406 ± 0.00002	0.9630 ± 0.0001	33.52 ± 0.02
	RadioUNet	0.02612 ± 0.00017	0.00382 ± 0.00005	0.9675 ± 0.0003	34.04 ± 0.08
	RadioET	**0.02446 ± 0.00046**	**0.00335 ± 0.00013**	**0.9693 ± 0.0011**	**34.61 ± 0.22**

TABLE III
TRAINING SETUP

Parameter	Value	Parameter	Value
Optimizer	AdamW	Batch Size	128
β_1, β_2	0.9, 0.95	Epochs	150
Weight Decay	10^{-4}	$\mathcal{N}_{\text{tx}}$	$\{2, 3, 4\}$
LR Schedule	Cosine	$\mathcal{M}_{\text{tx}}$	$(-6, 0)$ dB

Parameter	DAT-Unet	RadioUNet	RadioET
Learning rate	1×10^{-3}	5×10^{-4}	2×10^{-3}
DropPath Rate	0.1	–	–

minimizes the empirical risk, $\mathcal{L}_n^{(s)}(\widehat{F}_{\theta^*_{n,s}}) \leq \mathcal{L}_n^{(s)}(\widehat{F}_{\theta^*})$ for every training set. Since θ^* is fixed, its expected empirical risk equals $\delta(s)$. Taking expectations therefore yields the lower bound $\mathbb{E}[\mathcal{L}_n^{(s)}(\widehat{F}_{\theta^*_{n,s}})] \leq \delta(s)$. Also, $\delta(s) \leq \delta(n, s)$ because no data-dependent $\theta^*_{n,s}$ can attain population risk below the infimum. Combining, $\mathbb{E}[\mathcal{L}_n^{(s)}(\widehat{F}_{\theta^*_{n,s}})] \leq \delta(s) \leq \delta(n, s)$, so the training and test errors of the ERM bracket $\delta(s)$.

Fig. 4 shows the resulting sparsity scaling on logarithmic axes. The horizontal axis is the number of observed samples s, and the vertical axis is the full-map reconstruction MSE. The blue and orange curves show the training and test errors, respectively, and the shaded region between them gives the empirical bracket for the best-in-class population risk $\delta(s)$.

The red dashed curve shows the $\mathcal{O}(1/s)$ reference predicted by the approximation term in Corollary 1 (with $d = 2$). Even though the assumptions underlying the bound are only approximately satisfied in practice, the shaded region remains consistent with this reference over the range of s we examine. The remaining gap between training and test errors reflects finite-sample generalization effects. In particular, the training error measures the empirical risk minimized on the finite training set (up to optimization error), whereas the test error measures performance on unseen radio maps generated

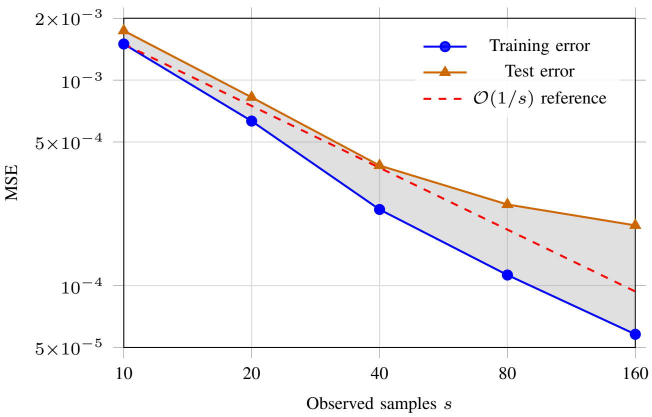

Fig. 4. Empirical sparsity scaling. Training and test MSE bracket the best-in-class population risk $\delta(s)$ (shaded band). The observed decay is consistent with the $\mathcal{O}(1/s)$ reference rate.

from the same benchmark distribution. As s increases, the approximation error decreases, but the finite training set can still limit how closely the learned reconstructor matches the best-in-class population risk, which explains why the train–test bracket remains visible.

VII. Discussion and Conclusion

We formulated sampling-based multi-transmitter RME as a sparse reconstruction problem and compared our empirical findings to a reference bound on the expected reconstruction risk. Empirically, RadioET achieves strong reconstruction performance on the multi-transmitter benchmark, and the error-decay experiments show monotone improvement with increasing s. The empirical decay is consistent with $\mathcal{O}(1/s)$, and establishing the rate under the relaxed assumptions of practical RME remains open.

While RadioET achieves notable gains in accuracy, these improvements introduce a trade-off with computational complexity. The energy-based block utilizes T iterative update steps per forward pass, and processing convolutional layers at high resolutions requires additional computational resources. Specifically, full training takes approximately 8h20m versus 2h40m for DAT-Unet and 2h10m for RadioUNet on $4\times$ RTX 6000 Ada GPUs under the identical pipeline.

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